BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 7 — Electrochemistry

ANSWER KEY (Version 1)

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

SECTION A — Multiple Choice Questions Answer Key (12 × 1 = 12 Marks)

Q1. In which change is nitrogen reduced? **√ C — N_2 to NH_3** **1 mark**

Reduction is gain of hydrogen. In $N_2 \rightarrow NH_3$, nitrogen gains hydrogen atoms, so it is **reduced**. Oxidation state of N changes from 0 to −3 (decrease = reduction). Options A and B show N gaining oxygen, which is oxidation.

Q2. Which is an example of oxidation? **√ B — Ag to Ag^+** **1 mark**

Oxidation is loss of electrons. $Ag \rightarrow Ag^+ + e^-$ shows silver losing one electron, so it is oxidized. Options A and C show gain of electrons (reduction). Option D shows $Fe^{3+} \rightarrow Fe^{2+}$, which is gain of electrons (reduction).

Q3. In $ZnO + H_2 \rightarrow Zn + H_2O$, which element is reduced? **√ B — ZnO** **1 mark**

ZnO is reduced because Zn in ZnO (oxidation state +2) loses oxygen and becomes Zn metal (oxidation state 0). The **oxidizing agent** is ZnO (the whole formula unit, not just the atom). H_2 is the reducing agent — it gains oxygen and is oxidized.

Q4. Oxidation state of Cr in $K_2Cr_2O_7$: **√ B — +6** **1 mark**

K = +1 (Group 1), so 2K gives +2. O = −2, so 7O gives −14. Let Cr = x; sum = 0: +2 + 2x + (−14) = 0 → 2x = 12 → x = **+6**. This is from Example 7.3 in the textbook (breathalyzer context). Option A is the total for both Cr atoms, not per atom.

Q5. Oxidation state of S in SO_4^{2-}: **√ C — +6** **1 mark**

In SO_4^{2-}, sum of oxidation states = −2 (the charge on the ion). O = −2, so 4O gives −8. Let S = x: x + (−8) = −2 → x = **+6**. A common distractor is +2 (Concept Assessment Exercise 7.4 in the textbook).

Q6. In $H_2S + Cl_2 \rightarrow 2HCl + S$, H_2S acts as: **√ A — Reducing agent** **1 mark**

H_2S is the **reducing agent**: S in H_2S has oxidation state −2 and changes to 0 in free S (oxidation, loss of electrons). Cl_2 is the oxidizing agent (Cl goes from 0 to −1, reduction). This reaction is from Review Question 6(d).

Q7. An oxidizing agent is a substance that: **√ B — gains electrons** **1 mark**

An **oxidizing agent** (electron acceptor) accepts electrons from another substance and is itself reduced in the process. Option A (loses electrons) describes a reducing agent. Options C and D describe the complementary oxygen/hydrogen definitions.

Q8. Formula of iron(III) chloride: **√ C — $FeCl_3$** **1 mark**

Iron(III) means Fe has oxidation state +3. Chloride ion is Cl^- (charge −1). To balance charges: 1 Fe^{3+} requires 3 Cl^-, giving **$FeCl_3$**. The textbook (section 7.3.1) states: "$FeCl_3$ is written as iron(III) chloride". Option B ($FeCl_2$) is iron(II) chloride.

Q9. Corrosion of iron (rusting) requires: **√ C — both oxygen and water** **1 mark**
The textbook states: "**Oxygen and water are necessary for iron to rust.**" The anodic region has less moisture (Fe oxidizes to Fe^{2+}) and the cathodic region has more moisture (O_2 is reduced using water). Neither alone is sufficient.

Q10. Cathodic protection of ships uses blocks of: **√ D — magnesium** **1 mark**
The textbook Fig. 7.1 shows **magnesium blocks** attached to ships. Magnesium is more active (reactive) than iron, so it acts as anode and sacrificially corrodes, protecting the iron hull. Copper and tin are less active than iron, so they cannot act as sacrificial anodes.

Q11. When acidified $KMnO_4$ meets a reducing agent, colour changes to: **√ B — colourless** **1 mark**
Activity 7.1 in the textbook: $KMnO_4$ is purple (MnO_4^-, Mn in +7 state). When reduced by a reducing agent, it becomes Mn^{2+} ions which are **colourless**. The change from purple to colourless confirms the presence of a reducing agent.

Q12. Oxidation number of a free element is always: **√ C — zero** **1 mark**
Rule 1 (Section 7.2.2): "**The oxidation state of any uncombined or free element is always zero.**" This applies to Zn, Na, H_2, S_8, etc. This is the most fundamental rule for assigning oxidation states.

SECTION B — Short Questions Answer Key (11 × 3 = 33 Marks)

2(i) — Oxidation and reduction: oxygen and electron definitions **3 marks**

Oxidation (oxygen): Oxidation is the **gain of oxygen** by a substance. Example: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$ — CO gains oxygen (oxidized). **1 mark**
Reduction (oxygen): Reduction is the **loss of oxygen** by a substance. In the same reaction, Fe_2O_3 loses oxygen (reduced). **1 mark**
Electron definitions: Oxidation = loss of electrons; Reduction = gain of electrons. Example: $Cu^{2+} + Zn \rightarrow Cu + Zn^{2+}$ — Zn loses $2e^-$ (oxidized); Cu^{2+} gains $2e^-$ (reduced). **1 mark**

OR

Oxidation (hydrogen): Oxidation is the **loss of hydrogen** by a substance. Example: $2C_2H_2 + 5O_2 \rightarrow 4CO_2 + 2H_2O$ — C_2H_2 loses hydrogen (oxidized). **1 mark**
Reduction (hydrogen): Reduction is the **gain of hydrogen** by a substance. Example: $N_2 + 3H_2 \rightarrow 2NH_3$ — N_2 gains hydrogen (reduced). **1 mark**
In the acetylene combustion reaction O_2 gains hydrogen to form H_2O, so O_2 is reduced. One accurate example with correct identification earns the third mark. **1 mark**

2(ii) — Rules for assigning oxidation states; N in HNO_3 **3 marks**

Four rules (Section 7.2.2):
Rule 1: Free/uncombined element = **0**. **½**
Rule 2: Monoatomic ion oxidation state = **its charge**. **½**
Rule 3: In a complex ion, sum of all oxidation states = **charge on ion**. **½**
Rule 4: In a neutral molecule/compound, algebraic sum of all oxidation states = **zero**. **½**

```
HNO3: H = +1, O = -2 (three O atoms, total = -6) Let N = x (+1) + x + 3(-2) = 0 1 + x - 6 = 0 x = +5
Oxidation state of N in HNO3 = +5
```

1 mark (for the correct working and answer)

OR

Free element rule: Oxidation state of any uncombined or free element = 0 (e.g. Na, Zn, S_8, H_2). **1 mark**
Monoatomic ion rule: The oxidation state = the charge on the ion, because a monatomic ion is formed by losing or gaining electrons from a single atom. **1 mark**
Na in Na^+ = **+1**; Fe in Fe^{3+} = **+3**. **1 mark**

2(iii) — Oxidation state of Cr in $K_2Cr_2O_7$ **3 marks**

```
K belongs to Group 1, so oxidation state of K = +1 There are 2 K atoms: contribution = 2(+1) = +2 [1 mark] O has oxidation state = -2 There are 7 O atoms: contribution = 7(-2) = -14 [1 mark] Let oxidation state of Cr = x There are 2 Cr atoms: contribution = 2x Applying Rule 4 (neutral compound, sum = 0): +2 + 2x + (-14) = 0 2x = 12 x = +6 Oxidation state of Cr in K2Cr2O7 = +6 [1 mark]
```

OR

Oxidation state of C in CO_3^{2-}:

```
CO3^2- is a polyatomic ion with charge = -2 (Rule 3) O has oxidation state = -2; there are 3 O atoms Contribution of O = 3(-2) = -6 [1 mark] Let oxidation state of C = x x + (-6) = -2 (sum = charge on ion = -2) x = -2 + 6 = +4 Oxidation state of C in CO3^2- = +4 [1 mark] (From Example 7.5 in the textbook)
```

1 mark for method (applying Rule 3 correctly)

2(iv) — Oxidation state of N in NO_2, N_2O, HNO_3 **3 marks**

```
(a) NO2: Let N = x; x + 2(-2) = 0 → x = +4 [1 mark] (b) N2O: Let each N = x; 2x + (-2) = 0 → 2x = 2 → x = +1 [1 mark] (c) HNO3: (+1) + x + 3(-2) = 0 → x - 5 = 0 → x = +5 [1 mark]
```

(This is from Review Question 7(b) in the textbook.)

OR

Oxidation state of S in H_2S, SO_2, $Na_2S_2O_3$:

```
(a) H2S: 2(+1) + x = 0 → x = -2 [1 mark] (b) SO2: x + 2(-2) = 0 → x = +4 [1 mark] (c) Na2S2O3: 2(+1) + 2x + 3(-2) = 0 2 + 2x - 6 = 0 2x = 4 → x = +2 [1 mark]
```

2(v) — Formulae of copper(II) sulphate, iron(II) and iron(III) chloride **3 marks**

(a) **Copper(II) sulphate:** Cu has oxidation state +2; SO_4^{2-} has charge −2. One Cu^{2+} balances one SO_4^{2-}. Formula: **$CuSO_4$**. **1 mark**

(b) **Iron(II) chloride:** Fe has oxidation state +2; Cl^- has charge −1. Need two Cl^- for one Fe^{2+}. Formula: **$FeCl_2$**. **1 mark**

(c) **Iron(III) chloride:** Fe has oxidation state +3; Cl^- has charge −1. Need three Cl^- for one Fe^{3+}. Formula: **$FeCl_3$**. **1 mark**

OR

(a) **Calcium chloride (Ca^{2+}, Cl^-):** Ca has charge +2, Cl has charge −1. Need 2 Cl^- for 1 Ca^{2+}. Formula: **$CaCl_2$**. **1.5 marks**

(b) **Aluminium sulphate (Al^{3+}, SO_4^{2-}):** Al has charge +3, SO_4^{2-} has charge −2. LCM of 3 and 2 = 6: need 2 Al^{3+} and 3 SO_4^{2-}. Formula: **$Al_2(SO_4)_3$**. Parentheses required for polyatomic ion. **1.5 marks**

2(vi) — Oxidizing and reducing agents; examples **3 marks**

Oxidizing agent: A substance that causes another substance to oxidize by taking electrons from it. It is an electron acceptor and is itself **reduced** in the process. **1 mark**

Reducing agent: A substance that causes another substance to be reduced by donating electrons to it. It is an electron donor and is itself **oxidized** in the process. **1 mark**

Common oxidizing agents (any three): oxygen (O_2), hydrogen peroxide (H_2O_2), chlorine (Cl_2), potassium permanganate ($KMnO_4$). **½**

Common reducing agents (any two): hydrogen gas (H_2), metal hydrides (e.g. $NaBH_4$), carbon monoxide (CO), metals such as Zn and Al. **½**

OR

Reaction: $2Na + Cl_2 \rightarrow 2NaCl$ **3 marks**

Assign oxidation states: Na starts at 0, becomes +1 in NaCl (increase → **oxidation**). Cl starts at 0 in Cl_2, becomes −1 in NaCl (decrease → **reduction**). **1 mark**

Na is the reducing agent (it is oxidized, it donates electrons). **1 mark**

Cl_2 is the oxidizing agent (it is reduced, it accepts electrons). Textbook (p.121): "Na is reducing agent as it is oxidized whereas Cl_2 is oxidizing agent as it is reduced." **1 mark**

2(vii) — $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$: identifying redox roles **3 marks**

```
Assign oxidation states: Fe2O3: Fe = +3, O = -2 CO: C = +2, O = -2 Fe: Fe = 0 CO2: C = +4, O = -2
```

(a) **Element oxidized:** Carbon (C) in CO goes from +2 to +4 (increase in oxidation state). **¾**

(b) **Element reduced:** Iron (Fe) in Fe_2O_3 goes from +3 to 0 (decrease in oxidation state). **¾**

(c) **Oxidizing agent:** Fe_2O_3 (contains the element being reduced — the whole formula unit). **¾**

(d) **Reducing agent:** CO (contains the element being oxidized — the whole formula unit). **¾**

(Note: "Oxidizing or reducing agent is the whole molecule or formula unit." — textbook p.122)

OR

Reaction: $WO_3 + 3H_2 \rightarrow W + 3H_2O$ (from Example 7.6):

```
W in WO3 = +6; W in W(metal) = 0 → W decreases: REDUCTION H in H2 = 0; H in H2O = +1 → H increases: OXIDATION
```

(a) H_2 is oxidized (H: 0 → +1). **¾** (b) WO_3 is reduced (W: +6 → 0). **¾** (c) Reducing agent: **H_2**. **¾** (d) Oxidizing agent: **WO_3**. **¾**

2(viii) — Activity 7.1: $KMnO_4$ and $FeSO_4$ **3 marks**

Observation: When a few drops of acidified $KMnO_4$ (purple) are added to $FeSO_4$ solution, the **purple colour is discharged** (turns colourless). **1 mark**

Explanation: $FeSO_4$ reduces $KMnO_4$. MnO_4^- ions (purple, Mn in +7 state) are reduced to Mn^{2+} ions (colourless, Mn in +2 state). The oxidation number of Mn changes from +7 to +2. **1 mark**

Oxidizing agent: $KMnO_4$ (it is reduced). **Reducing agent:** $FeSO_4$ (it is oxidized). **1 mark**

OR

Activity 7.2 observation: When H_2O_2 is added to potassium iodide (KI) solution, the solution turns **reddish-brown**, indicating the formation of iodine (I_2). **1 mark**

Equation: $H_2O_2 + 2KI + 2H^+ \rightarrow 2K^+ + I_2 + 2H_2O$ **1 mark**

Oxidizing agent: H_2O_2 (it is reduced; O in H_2O_2 goes from −1 to −2). KI is the reducing agent (I^- is oxidized to I_2, going from −1 to 0). **1 mark**

2(ix) — Corrosion definition and rusting half-reactions **3 marks**

Corrosion: Corrosion is a natural **electrochemical process** that occurs when a metal reacts with oxygen and moisture in the atmosphere, converting refined metals back to more stable metal oxides. **1 mark**

Anodic (oxidation) half-reaction (region with less moisture):

```
Fe(s) → Fe²⁺(aq) + 2e⁻ (oxidation)
```

1 mark

Cathodic (reduction) half-reaction (region with more moisture):

```
O₂ + 2H₂O + 4e⁻ → 4OH⁻ (reduction)
```

1 mark

OR

Why both oxygen and water are needed: The anodic process requires iron to oxidize: $Fe \rightarrow Fe^{2+} + 2e^-$. This occurs only in the presence of moisture (electrolyte). The cathodic process requires dissolved oxygen to accept electrons: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. Without oxygen there is no electron acceptor; without water there is no ionic conduction. Neither alone can sustain the electrochemical cycle. **2 marks**

Chemical formula of rust: $Fe_2O_3 \cdot xH_2O$ (hydrated iron(III) oxide). **1 mark**

2(x) — Galvanizing and electroplating **3 marks**

Galvanizing (Coating with Zinc): A clean iron sheet is dipped in a hot zinc chloride bath and heated, then rolled into a zinc bath and cooled. The zinc layer acts as a physical barrier AND as a sacrificial anode (zinc is more reactive than iron, so it corrodes preferentially). **1.5 marks**

Electroplating: An electrolytic process is used to deposit one metal on another metal. The object to be coated is made the cathode; the plating metal is made the anode in an electrolyte solution containing ions of the plating metal. Metal ions are reduced and deposited on the cathode. **1.5 marks**

OR

Cathodic (sacrificial) protection: The metal to be protected (e.g. iron) is made the **cathode** and connected to a more active metal such as magnesium or aluminium, which acts as the **anode**. **1 mark**

Why magnesium: Magnesium is more active (higher in the reactivity series) than iron, so it oxidizes preferentially. $Mg \rightarrow Mg^{2+} + 2e^-$. The Mg block corrodes (is sacrificed) instead of the iron hull. **1 mark**

Why the sacrificial metal must be more active than iron: Only a more active metal has a greater tendency to lose electrons. If it were less active than iron, iron would oxidize first and the protection would fail. Used in ships, underground pipes, tanks, and oil rigs. **1 mark**

2(xi) — Oxidation states in polyatomic ions SO_4^{2-}, PO_4^{3-}, NH_4^+ **3 marks**

```
(a) SO4^2-: charge = -2; O = -2; 4O total = -8 S + (-8) = -2 → S = +6 [1 mark] (b) PO4^3-: charge = -3; O = -2; 4O total = -8 P + (-8) = -3 → P = +5 [1 mark] (c) NH4^+: charge = +1; H = +1; 4H total = +4 N + (+4) = +1 → N = -3 [1 mark]
```

(These are from Concept Assessment Exercise 7.4 and 7.3 in the textbook.)

OR

Why Roman numerals are used: Roman numerals are used to indicate the oxidation states of elements in compounds when a metal exhibits **variable oxidation states**, particularly in transition metal compounds. **1 mark**

Examples from textbook: $CuSO_4$ is written as *copper(II) sulphate* (Cu = +2); $FeCl_2$ is *iron(II) chloride* (Fe = +2); $FeCl_3$ is *iron(III) chloride* (Fe = +3). Without Roman numerals, "iron chloride" would be ambiguous. **2 marks**

SECTION C — Long Questions Answer Key (4 × 5 = 20 Marks)

3(i) — Redox reactions: all three definitions + CuO/Zn ionic equation **5 marks**

(a) Definition of redox reactions: Redox reactions are chemical reactions in which oxidation and reduction occur simultaneously. A substance cannot be oxidized without another being reduced. **½**

Definition basis	Oxidation	Reduction
Oxygen transfer	Gain of oxygen	Loss of oxygen
Hydrogen transfer	Loss of hydrogen	Gain of hydrogen
Electron transfer	Loss of electrons	Gain of electrons
Oxidation number	Increase in oxidation number	Decrease in oxidation number

2.5 marks (1 mark per pair of definitions, up to 2.5) **(b) CuO + Zn → Cu + ZnO — ionic equation:** CuO and ZnO are ionic compounds. Rewrite as ions:

```
Molecular: CuO + Zn → Cu + ZnO Ionic: Cu²⁺ + O²⁻ + Zn → Cu + Zn²⁺ + O²⁻ O²⁻ ions appear on both sides (spectator ions); remove them: Net ionic: Cu²⁺ + Zn → Cu + Zn²⁺ [1 mark] Cu²⁺ gains 2 electrons → Cu (oxidation state +2 → 0): REDUCTION Zn loses 2 electrons → Zn²⁺ (oxidation state 0 → +2): OXIDATION [1 mark]
```

OR

(a) Four rules for assigning oxidation states:

1. Free/uncombined element = 0. **½**
2. Monoatomic ion = its ionic charge. **½**
3. Complex ion: sum of oxidation states = charge on the ion. **½**
4. Neutral molecule/compound: algebraic sum of all oxidation states = 0. **½**

Polyatomic ion rule (Rule 3) with example: In CO_3^{2-}, charge = −2. O = −2, so 3O = −6. C + (−6) = −2, C = +4. This is the carbonate ion worked in Example 7.5. **1 mark** **(b) Calculations:**

```
(i) H3BO3: 3(+1) + x + 3(-2) = 0 → 3 + x - 6 = 0 → x = +3 B in H3BO3 = +3 [½ mark] (ii) NH4⁺: charge = +1; H = +1; 4H = +4 N + 4 = +1 → N = -3 [½ mark] (iii) KMnO4: K=+1, O=-2; 4O=-8; Mn=x +1 + x + (-8) = 0 → x = +7 Mn in KMnO4 = +7 [1 mark]
```

3(ii) — Oxidizing and reducing agents: definition + three reactions 5 marks

(a) Definitions:

Oxidizing agent: A substance that causes another substance to be oxidized by accepting electrons from it. It is an **electron acceptor** and is itself reduced in the reaction. Common examples: O_2, H_2O_2, Cl_2, $KMnO_4$. 1 mark

Reducing agent: A substance that causes another substance to be reduced by donating electrons to it. It is an **electron donor** and is itself oxidized in the reaction. Common examples: H_2, CO, metals (Zn, Al). 1 mark **(b) Identification (1 mark per reaction):**

```
(i) 2S + Cl2 → S2Cl2 S: 0 → +1 (oxidized); Cl: 0 → -1 (reduced) Oxidizing agent: Cl2 Reducing agent: S [1 mark] (ii) 2Na + Br2 → 2NaBr Na: 0 → +1 (oxidized); Br: 0 → -1 (reduced) Oxidizing agent: Br2 Reducing agent: Na [1 mark] (iii) Mg + H2O → MgO + H2 Mg: 0 → +2 (oxidized); H: +1 → 0 (reduced) Oxidizing agent: H2O Reducing agent: Mg [1 mark]
```

OR

(a) $KMnO_4$ colour change test: Acidified $KMnO_4$ solution is purple due to MnO_4^- ions (Mn in +7 state). When added to a reducing agent, the colour changes from **purple to colourless**. MnO_4^- is reduced to Mn^{2+} ions (colourless), so the oxidation number of Mn decreases from +7 to +2. This colour change **confirms the presence of a reducing agent**. 2 marks **(b) Identification:**

```
(i) 2KI + Cl2 → 2KCl + I2 I: -1 → 0 (oxidized); Cl: 0 → -1 (reduced) Oxidizing agent: Cl2 Reducing agent: KI [1 mark] (ii) 2FeCl2 + Cl2 → 2FeCl3 Fe: +2 → +3 (oxidized); Cl2: 0 → -1 (reduced) Oxidizing agent: Cl2 Reducing agent: FeCl2 [1 mark] (iii) Mg + 2HCl → MgCl2 + H2 Mg: 0 → +2 (oxidized); H: +1 → 0 (reduced) Oxidizing agent: HCl Reducing agent: Mg [1 mark]
```

3(iii) — Formula of ionic compounds (three steps) + Roman numerals 5 marks

(a) Three steps to derive formula of an ionic compound:

Step 1: Identify the ions and their charges. ½

Step 2: Determine the simplest whole-number ratio of cations to anions that results in a **neutral compound** (total positive charge = total negative charge). ½

Step 3: Write the formula using the ratio from Step 2. Use parentheses for polyatomic ions when more than one is needed. ½ **Applying to aluminium sulphate (Al^{3+}, SO_4^{2-}):**

Step 1: Al^{3+} and SO_4^{2-}. Step 2: LCM of 3 and 2 = 6 → need 2 Al^{3+} and 3 SO_4^{2-}. Step 3: **$Al_2(SO_4)_3$**. Parentheses needed because more than one sulphate ion is required. 1.5 marks **(b) Formulae with oxidation states:**

```
(i) Copper(I) oxide: Cu = +1, O = -2 → Cu2O [½] (ii) Copper(II) oxide: Cu = +2, O = -2 → CuO [½] (iii)Iron(II) sulphate: Fe = +2, SO4 = -2 → FeSO4 [½] (iv) Manganese(IV) oxide: Mn = +4, O = -2 → MnO2 [½]
```

OR

(a) Roman numerals in chemical nomenclature: Roman numerals are used when a metal can exhibit **variable (multiple) oxidation states**, especially in transition metal compounds. Without them, the name would be ambiguous. Examples from textbook: $CuSO_4$ = copper(II) sulphate; $FeCl_2$ = iron(II) chloride; $FeCl_3$ = iron(III) chloride. 2 marks **(b) Oxidation state calculations:**

```
(i) NO2: x + 2(-2) = 0 → x = +4 (N = +4) [¾] (ii) H2SO4: 2(+1) + x + 4(-2) = 0 → 2+x-8=0 → x=+6 (S=+6) [¾] (iii)Cr2O3: 2x + 3(-2) = 0 → 2x=6 → x=+3 (Cr=+3) [¾] (iv) MnO4⁻: charge=-1; x+4(-2)=-1 → x-8=-1 → x=+7 (Mn=+7) [¾]
```

3(iv) — Corrosion: electrochemical nature, half-reactions, prevention methods **5 marks**

(a) Definition and electrochemical nature: Corrosion is a natural electrochemical process in which a metal reacts with oxygen and moisture in the atmosphere. It is electrochemical because it involves simultaneous **oxidation** (at the anodic region) and **reduction** (at the cathodic region), with electrons flowing between the two regions through the metal. **1 mark** **Anodic half-reaction** (less moisture region — oxidation):

```
Fe(s) → Fe²⁺(aq) + 2e⁻
```

1 mark **Cathodic half-reaction** (more moisture region — reduction):

```
O₂ + 2H₂O + 4e⁻ → 4OH⁻
```

Fe^{2+} ions flow to cathodic region and react with O_2 to form rust ($Fe_2O_3 \cdot xH_2O$). **1 mark** **(b) Four methods of preventing corrosion:**

1. **Coating with paint:** Creates a physical barrier between the metal and the environment (oxygen and moisture). Cheap and easily applied. **½**
2. **Alloying:** Adding chromium to iron forms stainless steel, which is highly resistant to corrosion (chromium forms a self-healing oxide layer). **½**
3. **Galvanizing:** Coating iron with zinc. Zinc acts both as a physical barrier and as a sacrificial anode. **½**
4. **Electroplating:** Using an electrolytic process to deposit a corrosion-resistant metal (e.g. chromium, tin) on the iron surface. **½**

OR

(a) Cathodic (sacrificial) protection — detailed explanation: The metal to be protected (iron) is made the **cathode** by connecting it to a more active metal (magnesium or aluminium), which acts as the **anode**. The more active metal has a greater tendency to lose electrons, so it oxidizes preferentially:

```
Mg → Mg²⁺ + 2e⁻ (Mg block corrodes; iron is protected)
```

Why magnesium over copper: Magnesium is higher in the reactivity series (more active) than iron. Copper is *less* active than iron, so connecting copper would make iron the anode and it would corrode faster. **1 mark**
Structures: Underground pipes, storage tanks, ship hulls, oil rigs, and marine structures. The textbook shows magnesium blocks attached below the waterline of ships (Fig. 7.1). **2 marks** **(b) Tinning vs Galvanizing comparison:**

Heading	Tinning	Galvanizing
(i) Metal used	Tin (Sn)	Zinc (Zn)
(ii) Process	Clean iron sheet dipped in molten tin, then passed through hot rollers	Clean iron sheet dipped in hot zinc chloride bath, then rolled into zinc bath and cooled
(iii) Application	Food cans (tin is stable and non-toxic)	Roofing sheets, water tanks, pipes, bridges

2 marks